

DISTRIBUTED SYSTEMS

[Revised Credit System]
(Effective from the academic year 2018-19)
SEMESTER - VI

Subject Code	CSE3251	IA Marks	50
Number of Lecture Hours/Week	04	Exam Marks	50
Total Number of Lecture Hours	48	Exam Hours	03
CREDITS - 04			
Course objectives: This course will enable students to understand • the different architectures of distributed systems including the Hadoop Ecosystem • how various processes synchronize with each other • the different types of interprocess communication • the different methods for naming • replication and mechanisms for maintaining consistency among these replicas			
Module – 1			Teaching Hours
Introduction: What is a distributed system? Design Goals. Architecture: Architectural Styles, Middleware Organization, System Architecture, Example Architectures Text 1: Ch 1: 1.1, 1.2 Text 1: Ch 2			6 Hours
Module -2			Teaching Hours
Meet Hadoop: Data!, Data Storage and Analysis, Querying All Your Data, Beyond Batch, Comparison with Other Systems, A Brief History of Apache Hadoop MapReduce: A Weather Dataset, Analyzing the Data with Hadoop, Scaling Out, Hadoop Streaming The Hadoop Distributed Filesystem: The Design of HDFS, HDFS Concepts, The Command-Line Interface, Hadoop Filesystems, Data Flow YARN: Anatomy of a YARN Application Run, YARN Compared to MapReduce 1, Scheduling in YARN How MapReduce Works: Anatomy of a MapReduce Job Run, Failures, Shuffle and Sort, Task Execution			6 Hours

Text 2: Ch 1, Ch 2, Ch 3, Ch 4, Ch 7	
Module -3	
Coordination: Clock synchronization, Logical clocks, Mutual exclusion, Election algorithms Text 1: Ch 6: 6.1, 6.2, 6.3, 6.4	10 Hours
Module – 4	
Communication: Foundations, Remote procedure call, Message-oriented communication, Multicast communication Text 1: Ch 4	8 Hours
Module-5	
Naming: Names, Identifiers and Addresses, Flat naming, Structured naming Text 1: Ch 5: 5.1, 5.2, 5.3	8 Hours
Module-6	
Consistency & Replication: Introduction, Data-centric consistency models, Client-centric consistency models, Replica management, Consistency protocols Text 1: Ch 7: 7.1 – 7.5	10 Hours
Course outcomes:	
After studying this course, students will be able to: 1. explain the different architectures of distributed systems including the Hadoop Ecosystem 2. figure out the mechanism of synchronization of processes 3. explain interprocess communication between processes 4. explain the working of a naming service in a distributed system 5. comprehend replication and mechanisms for maintaining consistency	
Text Books: 1. Distributed Systems – Maarten van Steen and Andrew S. Tanenbaum, 3rd edition, Version 3.01, 2017 2. Hadoop The Definitive Guide, Tom White, 4th Edition, Oreilly Publication.	

Reference Books:

1. Distributed Systems - Coulouris G., Dollimore J., and Kindberg T., Pearson, 4th edition, 2009.
2. Hadoop in Action - Chuck Lam, Manning Publications Co.
3. Ajay D. Kshemkalyani, and Mukesh Singhal, Distributed Computing: Principles, Algorithms, and Systems, Cambridge University Press; Reissue edition, March 2011.
4. Mei- Ling Liu, *Distributed Computing: Principles and Application*, Pearson Education, Inc. New Delhi. 2004.